

P.3: Integrating Algorithms into Commission Redistricting: The Iowa Model and Three Integration Modes

Giovanni Della-Libera

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Abstract

Fourteen states currently use some form of independent or bipartisan commission to draw congressional or state legislative districts. None currently requires algorithmic redistricting. This paper analyzes three modes for integrating the **bisect** recursive bisection algorithm into commission redistricting processes: Mode (a), algorithm-as-sole-mapmaker (the commission approves or rejects the algorithmic output); Mode (b), algorithm-as-first-draft (the algorithm produces a baseline that the commission modifies under documented criteria); and Mode (c), algorithm-as-audit (the commission draws maps independently, then uses the algorithm to verify compliance with stated criteria). We find that Mode (b) is most feasible for existing commissions: it preserves the commission’s deliberative function, provides a defensible neutral starting point, and creates a verifiable record of departures. We develop the Iowa model as the strongest existing analogue to algorithmic commission redistricting: Iowa’s Legislative Services Agency (LSA) draws maps using strictly enumerated criteria with no partisan data, and the legislature votes up or down on the LSA’s proposals without amendments. The **bisect** platform extends the Iowa model by replacing the LSA’s manual map-drawing with deterministic algorithmic output, eliminating the residual human manipulation space in criteria-only approaches. We provide a detailed Commission Adjustment Protocol specifying the procedures, documentation requirements, and certification steps for Mode (b) integration.

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1 Introduction

The 14 states that use independent or bipartisan redistricting commissions represent the most politically established infrastructure for neutral redistricting in the United States. California, Colorado, Michigan, Arizona, Hawaii, Idaho, Montana, New Jersey, New York, Ohio, Virginia, Washington, and Alaska all use commissions for at least one chamber; Iowa uses a non-commission variant (the Legislative Services Agency model) that achieves similar neutrality through different institutional design.

Commission redistricting has significant advantages over legislative redistricting: commissions are structurally independent, typically required to hold public hearings, and must satisfy enumerated criteria rather than partisan objectives. But commissions are not gerrymander-proof. Even well-designed commissions can produce maps that favor one party through the cumulative effect of individually defensible line-drawing decisions. The California CCRC’s 2021 congressional maps, for example, were criticized by some analysts for producing more Democratic-leaning seats than a neutral algorithm would generate—not because the commission acted with partisan intent, but because the commission’s good-faith application of criteria (particularly community of interest) has distributional consequences that are not transparent to the commission.

The **bisect** platform addresses this gap. By providing a deterministic baseline that any person can verify, it makes the distributional consequences of commission criteria explicit: the difference between the algorithmic baseline and the commission’s final map is the quantifiable effect of the commission’s discretionary choices. This transparency serves two goals. It holds commissions accountable by making manipulation visible. And it protects commissions from unfair criticism by proving that departures from neutrality are VRA-required or otherwise legally justified.

This paper is addressed to redistricting commissioners, state administrators, and commission legal counsel. Section 2 examines the Iowa model as the strongest existing analogue. Section 3 describes three integration modes. Section 4 provides a detailed Commission Adjustment Protocol for Mode (b) integration. Section 5 concludes.

2 The Iowa Model as Precedent

Iowa is the United States’ most successful example of non-partisan legislative redistricting. Since 1980, Iowa’s congressional and state legislative districts have been drawn by the Legislative Services Agency (LSA), a nonpartisan staff office of the Iowa Legislature, subject to legislative approval. The process is constrained by Iowa Code §42.4, which specifies the

criteria in strict priority order and prohibits the LSA from considering any political data.

2.1 How the Iowa Model Works

After each census, the LSA draws congressional and state legislative maps satisfying the following criteria in order:

1. Population equality (congressional districts must satisfy strict one-person, one-vote; state districts allow up to 1% deviation)
2. Contiguity
3. Compactness (explicitly defined as length-width ratio minimization)
4. Political subdivision respect (county and municipal boundary preservation)

The LSA draws maps without access to voter registration data, party registration, election results, or incumbency information. The LSA staff are career civil servants who have no electoral stake in the outcome.

The legislature receives the LSA map and votes it up or down without amendment. If the legislature rejects the first map, the LSA draws a second map on the same criteria. If the second map is rejected, the LSA draws a third map. If the third map is rejected, the courts take over. In practice, Iowa has never reached the third map: the first or second LSA map has always been adopted. The legislature's ability to reject maps without power to amend them creates a take-it-or-leave-it dynamic that substantially limits gaming.

2.2 Iowa's Track Record

Iowa's redistricting has been remarkably controversy-free compared to other states. Since 1980:

- No Iowa redistricting has been struck down in federal court for constitutional violations.
- Iowa's congressional districts consistently rank among the most compact in the nation (mean Polsby-Popper score of approximately 0.38, compared to a national congressional average of 0.25).
- Iowa's maps have been drawn and adopted within approximately three months of receiving census data, far faster than the average commission or legislative process.

- Iowa spent approximately \$3 million on the 2021 redistricting cycle, including legislative hearings and legal review—a fraction of the \$20M+ spent by California and the \$40M+ spent on redistricting litigation in states like North Carolina.

2.3 Iowa’s Limitations

The Iowa model has three limitations relevant to algorithm integration.

First, the LSA’s “non-partisan” process still involves human judgment: mapmakers make decisions about how to apply the criteria (which subdivision boundary to preserve when preserving all of them is geometrically impossible; which compact shape to prefer when multiple compact shapes satisfy the compactness criterion). These decisions are not transparent and are not verifiable.

Second, Iowa does not have VRA obligations comparable to states with large minority populations. Iowa’s racial demographics—85% non-Hispanic white—mean the VRA rarely requires majority-minority districts. In states with substantial minority populations (Alabama, Georgia, Texas), the Iowa model as written would be legally insufficient without an explicit VRA criterion and a documented modification process.

Third, Iowa’s legislative approval-or-reject structure is not replicable in states where the legislature is strongly motivated to gerrymander. The threat of court takeover is credible in Iowa because the courts would likely produce maps similar to the LSA’s. In states where courts have not historically been active in redistricting, the credibility of the court-takeover threat is lower.

2.4 The Algorithmic Extension of the Iowa Model

The `bisect` platform extends the Iowa model by replacing the LSA’s manual map-drawing with deterministic algorithmic output. Under algorithmic Iowa-style redistricting:

```
bisect build iowa_congressional --year 2030 --workers 8
bisect label-verify iowa_congressional --year 2030
```

The verification step produces a manifest confirming that the map was drawn from public census data, without partisan inputs, with a compactness-optimizing algorithm. The LSA (or equivalent body in other states) retains authority to document VRA modifications using the Commission Adjustment Protocol in Section 4. The legislature receives the algorithmic output plus the VRA modification documentation and votes up or down.

This combination eliminates the residual human manipulation space in the Iowa model

while preserving the institutional structure—bipartisan staff, legislative approval, criteria-based constraint—that has made Iowa’s process politically legitimate.

3 Three Integration Modes

We define three distinct modes for integrating the `bisect` algorithm into commission redistricting processes, each representing a different allocation of authority between the algorithm and the commission.

3.1 Mode (a): Algorithm-as-Sole-Mapmaker

In Mode (a), the commission’s role is limited to approving or rejecting the algorithmic output. The algorithm runs on census data and produces a map. The commission may, on a supermajority vote, reject the map and instruct the algorithm to re-run with different parameters. But the commission may not modify individual district lines: the algorithm’s output is the map, subject only to rejection.

Advantages. Mode (a) provides the strongest protection against partisan manipulation: because the commission cannot amend individual lines, there is no mechanism by which commissioners can introduce partisan bias through the adjustment process. The algorithm’s structural neutrality is fully preserved.

Disadvantages. Mode (a) is politically difficult: commissioners, advocacy groups, and legislators who believe the commission should exercise judgment (for VRA compliance, community of interest, and political subdivision integrity) will resist a mode that eliminates commission discretion. More substantively, VRA compliance sometimes requires that specific minority communities be included in a single district, and an algorithm constrained only to population balance and compactness may not always achieve this. A pure Mode (a) design requires a VRA-override mechanism that restores the ability to modify specific lines, which substantially weakens the mode’s structural protection.

Best suited for. States without substantial minority populations requiring VRA modification (analogous to Iowa); states where the legislature retains a post-commission approval role (the legislature’s veto constrains the commission’s discretion); courts ordering remedial maps (P.4 develops court-ordered Mode (a) in detail).

3.2 Mode (b): Algorithm-as-First-Draft

In Mode (b), the algorithm produces a baseline map that the commission modifies under documented criteria. The commission retains full authority to adjust individual district lines,

but must document each departure from the algorithmic baseline with a written justification. For VRA-motivated adjustments, the documentation must satisfy the *Gingles* three-prong framework. For other adjustments, the documentation must identify the statutory criterion satisfied (political subdivision integrity, community of interest, etc.) and the specific boundary change made.

Advantages. Mode (b) preserves the commission’s deliberative function—the human judgment that voters and advocacy groups value—while providing a verifiable baseline that makes the distributional consequences of commission decisions transparent. The documentation requirement creates a record that courts can evaluate, limiting the commission’s discretion to adjustments grounded in statutory criteria. The algorithm’s integrity is maintained as a verifiable reference point even when the commission departs from it.

Disadvantages. Mode (b) requires more administrative capacity than Mode (a) (the commission must manage the documentation process) and more than Mode (c) (the commission must consult the algorithmic baseline throughout its deliberations, not merely at the end). The documentation requirement may slow the commission’s process if commissioners are unfamiliar with the *Gingles* framework and other legal standards.

Best suited for. States with independent commissions and substantial minority populations requiring VRA compliance (California, Michigan, Colorado, Arizona); commissions with legal counsel competent in VRA; situations where the commission’s public legitimacy depends on demonstrable deliberation.

3.3 Mode (c): Algorithm-as-Audit

In Mode (c), the commission draws maps through its normal process, then uses the **bisect** platform to verify that the final map satisfies the criteria the commission claims to have applied. The algorithm serves as a post-hoc validity check rather than a starting point. If the algorithmic audit reveals that the final map departs significantly from the criteria-optimal baseline, the commission must explain the departure.

Advantages. Mode (c) requires no change to the commission’s existing process. It imposes minimal administrative burden during the map-drawing phase. It provides an audit mechanism that can detect if the commission’s stated justifications are inconsistent with the algorithmic analysis (e.g., if the commission claims to have optimized compactness but the algorithmic audit shows multiple districts with substantially lower compactness than the algorithm achieves).

Disadvantages. Mode (c) does not constrain the commission during map-drawing; it can only expose inconsistencies after the fact. A commission that draws a partisan map can

satisfy the audit by providing pretextual explanations for each departure. The audit’s force depends on the threat of litigation or public criticism—it is a transparency mechanism, not a structural constraint. Mode (c) is the weakest of the three modes for preventing manipulation.

Best suited for. States where political constraints prevent requiring algorithmic baselines; commissions where algorithm integration is being piloted before full Mode (b) adoption; external audits by good-government organizations reviewing commission maps after adoption.

3.4 Feasibility Assessment

Mode (b) is our recommended integration mode for existing commissions. It achieves the most important goal—a verifiable, neutral baseline that constrains manipulation while preserving legitimate commission discretion—with a feasible administrative process that existing commissions can implement without legislative action.

The California Citizens Redistricting Commission could adopt Mode (b) through a standing rule or operating procedure, without a ballot initiative, by designating the algorithmic baseline as the starting point for its map-drawing process and adopting the Commission Adjustment Protocol in Section 4. The amendment to California Government Code described in P.2, Section 4 would formalize this approach constitutionally.

4 The Commission Adjustment Protocol

The Commission Adjustment Protocol (CAP) specifies the procedures, documentation requirements, and certification steps for Mode (b) integration of the `bisect` platform into commission redistricting. It is designed to be adopted as a commission operating rule; it does not require statutory change.

4.1 Step 1: Baseline Generation

The commission’s technical staff runs the `bisect` platform to generate the algorithmic baseline map:

```
bisect fetch --year 2030 --workers 8
bisect build [state]_baseline --year 2030 --workers 8
bisect label-analyze [state]_baseline --year 2030 --types all
bisect label-report [state]_baseline --year 2030 --format html json
```

The baseline map and its reproducibility manifest are published on the commission’s public website within 30 days of receiving the decennial census redistricting data. The

manifest records the binary version, census data release, TIGER adjacency file hash, balance tolerance, and seed. Any member of the public may verify the baseline by running the platform with the recorded parameters.

4.2 Step 2: Public Review of Baseline

The commission holds at least three public hearings to present the algorithmic baseline and receive comment. Comments specifically addressing departures from the baseline—requests that the commission modify specific district lines for VRA, community of interest, or political subdivision reasons—are logged in a public comment register. The commission designates a specific deadline for departure requests.

4.3 Step 3: Departure Deliberation

For each requested or proposed departure from the baseline, the commission records a Departure Finding that includes:

1. **Affected districts:** GEOID-level identification of the census tracts to be moved from one district to another.
2. **Criterion invoked:** The statutory criterion (VRA compliance, political subdivision integrity, community of interest, compactness, contiguity) that justifies the departure.
3. **VRA departures:** For departures motivated by Section 2 of the Voting Rights Act, the Departure Finding must satisfy the *Gingles* three-prong framework [[Supreme Court of the United States, 1986](#)]:
 - (i) The minority group is “sufficiently large and geographically compact” to constitute a majority in a single-member district;
 - (ii) The minority group is “politically cohesive”; and
 - (iii) The majority votes sufficiently as a bloc “to enable it to defeat the minority’s preferred candidate.”

For coalition-district claims, the finding must also identify the specific coalition communities and the evidence of their combined political cohesion, consistent with *Allen v. Milligan*, 599 U.S. 1 (2023).

4. **Population impact:** The departure must maintain population balance within the commission’s adopted tolerance (typically $\pm 0.5\%$ of ideal district population).

5. **Compactness impact:** The departure finding must report the change in Polsby-Popper score for each affected district.

4.4 Step 4: Re-Certification

After recording all departures, the commission’s technical staff re-runs the **bisect** population-balance certification:

```
bisect label-analyze [state]_final --year 2030 --types population
bisect label-verify [state]_final --year 2030
```

The verification command confirms that the final map (incorporating all departures) satisfies population balance. Any district that fails the population balance check must be re-drawn before adoption. The re-certification is published as a supplement to the original manifest.

4.5 Step 5: Final Publication

The commission publishes the following artifacts before adopting the final map:

- The algorithmic baseline manifest (**manifest.json** from the baseline run)
- The Departure Registry (all Departure Findings, with evidence supporting VRA prongs)
- The final map’s population balance certification
- A difference file identifying each census tract moved from the baseline assignment to the final assignment

These artifacts are retained for 12 years (one and a half redistricting cycles) to support any litigation challenging the final map.

4.6 Anticipated Legal Scrutiny

Courts reviewing commission maps adopted under the CAP should evaluate:

VRA departures: Whether the Departure Finding identifies a specific minority community, satisfies the *Gingles* prongs with record evidence, and makes a narrowly tailored modification consistent with *Miller v. Johnson*, 515 U.S. 900 (1995)’s requirement that race not be the predominant factor.

Non-VRA departures: Whether the Departure Finding identifies a specific statutory criterion and whether the modification is consistent with the commission’s adopted criteria

hierarchy. Courts should apply deferential review to non-VRA departures, as commission discretion in applying criteria is the purpose of the commission structure.

Baseline integrity: Whether the algorithmic baseline manifest is reproducible. Any party claiming the baseline was manipulated must demonstrate that running the **bisect** platform with the recorded parameters produces a different map. If verification succeeds, the baseline’s integrity is established.

The CAP’s documentation requirements are designed to satisfy the record-keeping standards of *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993), applicable to expert testimony, and the administrative record requirements of *Motor Vehicle Manufacturers Association v. State Farm Mutual Automobile Insurance Co.*, 463 U.S. 29 (1983), applicable to formal agency proceedings.

5 Conclusion

Algorithmic redistricting and commission redistricting are not alternatives. They are complements. The algorithm provides what human commissioners cannot: a verifiable, deterministic baseline produced from public data without partisan input. The commission provides what the algorithm cannot: legitimate deliberation over VRA requirements, community of interest considerations, and political subdivision integrity—judgments that require contextual knowledge and public accountability.

Mode (b) integration—algorithm-as-first-draft—achieves the most favorable balance between these contributions. It preserves commission authority over VRA and other legally required adjustments while making those adjustments transparent and verifiable. The Commission Adjustment Protocol specifies the process in sufficient detail to be implementable by any existing commission without statutory change; its adoption as a commission operating rule is a feasible near-term step for commissions in California, Colorado, Michigan, Arizona, and other states with independent redistricting authority.

The Iowa model demonstrates that non-partisan criteria-based redistricting can produce durable, cost-effective maps that command public legitimacy. Iowa’s limitation—the residual human judgment in the LSA’s map-drawing—is precisely what the **bisect** platform eliminates. An algorithmic Iowa-style process would retain Iowa’s institutional strengths (nonpartisan staff, legislative approval-or-reject, criteria-based constraint) while replacing manual map-drawing with verifiable algorithmic output.

For redistricting commissioners and administrators reading this paper, the practical implication is direct: the **bisect** platform can be run today on 2020 census data to produce baseline maps for any state, demonstrating its functionality before the 2030 redistricting

cycle. States that begin experimenting with Mode (b) integration in 2026–2028 will be in the strongest position to implement it effectively in 2031. The investment in training, process design, and public communication can be spread over five years rather than compressed into the six months after census data release.

The political barrier to Mode (b) adoption is modest: unlike federal legislation (which requires congressional majorities) or ballot initiatives (which require statewide campaigns), a commission operating rule requires only a commission vote. Commissioners appointed to produce neutral maps should find a Mode (b) algorithmic baseline—which provides the most transparent possible demonstration that the commission’s maps are not gerrymandered—congruent with their institutional purpose.

References

Supreme Court of the United States. *Thornburg v. Gingles*, 478 U.S. 30, 1986.